Make sense of a chance forecast

Read a probability as a claim about a defined event, then judge a record rather than one outcome.

Suggested teaching time: 30 min | Preparation: about 5 min

Introductory | Secondary and adult learners | Individual writing, pairs, and class discussion

Prerequisites: Learners can read percentages as numbers out of 100.

Students will be able to

- Name the event and time covered by a forecast.
- Translate a percentage into approximate outcomes across comparable cases.
- Explain why one outcome does not settle whether a probability is well judged.

Prepare and gather

- One student handout per learner, or display the activities and use paper.
- Pens or pencils; the instructor guide stays separate from student materials.
- A basic calculator is optional for the exit-ticket percentage.
- Read all three scenarios and their model answers.
- Print the handout or display one activity at a time. Explain that all scenarios and figures are invented for reasoning practice.

Teaching sequence

5 min	**Translate a percentage.** Learners complete the bus warm-up. Check that they interpret 25% as a chance of the defined late-arrival event.
10 min	**Read the rain forecast.** Pairs complete the main task. Ask them to name the rain threshold and time window before discussing expected counts.
8 min	**Judge the record.** Use the discussion prompts. Emphasize that calibration concerns many forecasts, and exact counts can vary.
7 min	**Check a delivery forecast.** Learners complete the delivery exit ticket independently. Compare their answer with the criteria and model answer.

Timings are suggested starting points, not measured classroom completion times.

Warm-up / A chance of arriving late

Student task and instructor answer

Explain these words as needed

Probability: A stated chance that a defined event will occur.

Calibration: A match between stated chances and outcome frequencies across many comparable forecasts.

Comparable forecasts: Forecasts made for similar events, conditions, and time horizons.

Scenario

Invented scenario: A travel app gives tomorrow's bus a 25% chance of reaching the terminal after 9 a.m. Imagine 100 comparable forecasts made one day ahead at 25% by a well-calibrated app: its chances match long-run frequencies.

Student output: Write two approximate counts and one sentence about the chance of tomorrow being late.

1. State about how many of the 100 buses should arrive after 9 a.m. and how many should not.

2. Explain why a 25% forecast does not rule out a late bus tomorrow.

Student success check: Keep the event “after 9 a.m.” clear; do not turn “unlikely” into “impossible”.

Model answer

About 25 buses should arrive after 9 a.m. and about 75 should not; exact counts can vary. A late bus tomorrow is possible because a 25% chance includes that outcome.

Why this meets the task: It translates the chance into an approximate frequency while preserving uncertainty about one journey.

Main activity / What does 70% rain mean?

Student task and instructor answer

Scenario

Invented scenario: A weather app assigns a 70% chance of measurable rain tomorrow. Here, measurable rain means at least 1 mm at the town weather station during that calendar day. Imagine 100 comparable forecasts, each issued one day ahead at 70%, from a well-calibrated app: its stated chances match long-run frequencies.

Student output: Write two approximate counts, an explanation of one dry day, and a plan for checking the forecast record.

1. Write approximately how many of those 100 days should have measurable rain and how many should not.
2. Explain why a dry tomorrow would not by itself refute a 70% forecast.
3. Describe the record you would inspect to judge whether the app's 70% forecasts are well calibrated.

Student success check: Do not treat 70% as a guarantee, 70% of the town, or 70% of the day.

Model answer

About 70 days should have measurable rain and about 30 should not; exact counts can vary. A dry tomorrow is allowed by the 30% chance of no measurable rain. Inspect many comparable forecasts recorded in advance at 70%, include both wet and dry days, and compare their rain frequency with 70%.

Why this meets the task: It identifies the event and evaluates a chance claim across a complete record instead of treating one result as decisive.

Exit ticket / A delivery forecast record

Student task and instructor answer

Scenario

Invented scenario: A courier predicts an 80% chance that a parcel will arrive by 6 p.m. tomorrow. Its record contains 200 comparable one-day-ahead forecasts made at 80%; 160 parcels arrived by 6 p.m. and 40 did not.

Student output: Write a frequency calculation and two sentences judging the record and a late parcel.

1. Calculate the on-time percentage: on-time parcels / all parcels x 100.
2. State whether this record matches the stated 80% chance, without claiming it guarantees future accuracy.
3. Explain why one late parcel would not by itself show an 80% forecast was wrong.

Student success check: Include all 200 forecasts and preserve the possibility of a late parcel.

Model answer

160/200 x 100 = 80%. The recorded frequency matches the stated 80% chance in this set, but does not guarantee future calibration. One late parcel is compatible with the remaining 20% chance of missing the deadline.

Why this meets the task: It uses the complete record and separates agreement in this set from certainty about any particular future delivery.

Check understanding

Mark each criterion met or needs another attempt. Accept equivalent wording; assess the reasoning, not a memorized phrase.

- The forecast is tied to its stated event, threshold, and time period.
- The frequency calculation includes all relevant cases and is correct.
- The interpretation preserves uncertainty and distinguishes a record from a single result.

Discussion and adaptations

Use the supplied guidance to keep the lesson focused on its reasoning skill.

Discussion prompts

Can a well-judged forecast assign 70% and still be followed by a dry day?

Listen for: Yes. The forecast explicitly leaves a 30% chance of no measurable rain. A single outcome cannot show whether 70% was the right degree of confidence.

Why keep unsuccessful forecasts in the record?

Listen for: Selecting only forecasts followed by rain would distort the measured frequency. Record the prediction before the event and retain every relevant outcome.

Common confusions

More likely means certain. A chance above 50% still allows the other outcome.

A good 70% forecast produces exactly 70 wet days in every set of 100. Calibration describes a long-run pattern. Finite sets can vary around that frequency.

Matching one percentage proves a system is useful for every forecast. Inspect other probability groups, conditions, and periods too. One matching group establishes only that limited agreement.

Adapt the lesson

More support: Use 100 boxes and shade 70 for measurable rain and 30 for no measurable rain. Ask whether choosing an unshaded box was ruled out. Expected answer: no; it represents an allowed outcome.

More challenge: Ask what 120 on-time deliveries among the same 200 forecasts would imply. Expected calculation: 120/200 x 100 = 60%, below the predicted 80%. That mismatch warrants checking calibration; it does not prove deliberate dishonesty.

Next practice

https://xixteen.com/learners/?practice=probability#workbench

Who Deserves Your Trust? Ask What They Can Predict:
https://xixteen.com/articles/prediction-markets-and-earned-trust/